4 Zadaca - Operacijski sustavi

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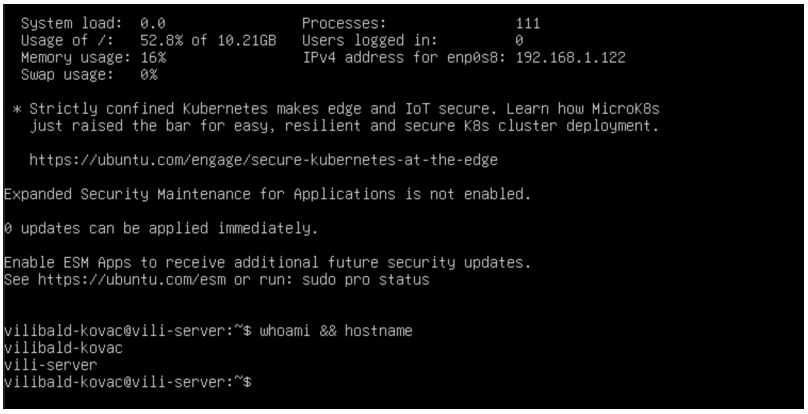
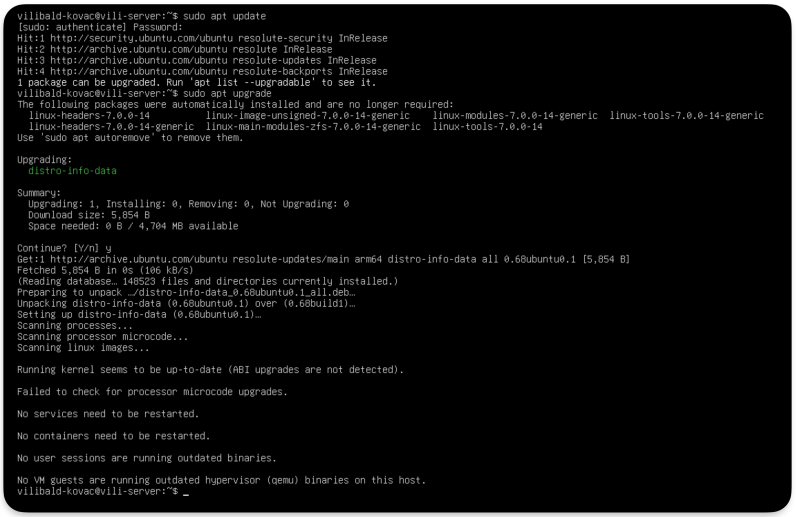
Instalacija

1. **Preuzmite i instalirajte** VirtualBox na svoje računalo. ✅
2. **Preuzmite i instalirajte** Ubuntu Server (LTS verziju) te **izradite novi virtualni stroj**.
   Tijekom instalacije Ubuntu Servera, za `username` unesite svoje ime i prezime u formatu: `ime.prezime`. ✅
3. Nakon instalacije, **napravite *screenshot* početnog zaslona** Ubuntu Servera na kojem se jasno vidi vaše korisničko ime i naziv virtualnog stroja.



2. **Za svaki od sljedećih zadataka napravite odgovarajući *screenshot*:**

- Ažurirajte lokalnu listu dostupnih paketa i verzija te nadogradite sve pakete na najnovije verzije.



- Instalirajte paket `openssh-server`, pokrenite SSH poslužitelj i provjerite njegov status.

```
Preparing to unpack .../ncurses-term_6.6+20251231-1_all.deb...
Unpacking ncurses-term (6.6+20251231-1)...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a10.2p1-2ubuntu3.2_arm64.deb...
Unpacking openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a10.2p1-2ubuntu3.2_arm64.deb...
Unpacking openssh-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu4_all.deb...
Unpacking ssh-import-id (5.11-0ubuntu4)...
Setting up openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up libwrap0:arm64 (7.6.q-36build2)...
Setting up ncurses-term (6.6+20251231-1)...
Setting up openssh-server (1:10.2p1-2ubuntu3.2)...
Creating config file /etc/ssh/sshd_config with new version
Creating user `sshd' (sshd user) with UID 984 and GID 65534.
Created symlink '/etc/systemd/system/sockets.target.wants/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.requires/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2)...
Scanning processes...
Scanning processor microcode...
Scanning linux images...

Running kernel seems to be up-to-date (ABI upgrades are not detected).

Failed to check for processor microcode upgrades.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vilibald-kovac@vili-server:~$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: • ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
vilibald-kovac@vili-server:~$ _
```

```
vilibald-kovac@vili-server:~$ sudo systemctl start ssh
[sudo: authenticate] Password:
vilibald-kovac@vili-server:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Sat 2026-05-09 11:07:50 UTC; 17min ago
 Invocation: 11ffd8fe6e124752a5e1da6406d2ebc0
TriggeredBy: • ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 11424 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 11426 (sshd)
      Tasks: 1 (limit: 950)
     Memory: 1.1M (peak: 3.2M)
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─11426 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 09 11:07:50 vili-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 09 11:07:50 vili-server sshd[11426]: Server listening on 0.0.0.0 port 22.
May 09 11:07:50 vili-server sshd[11426]: Server listening on :: port 22.
May 09 11:07:50 vili-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
vilibald-kovac@vili-server:~$
```

- Pronađite IP adresu virtualnog stroja i provjerite koji su mrežni portovi otvoreni.

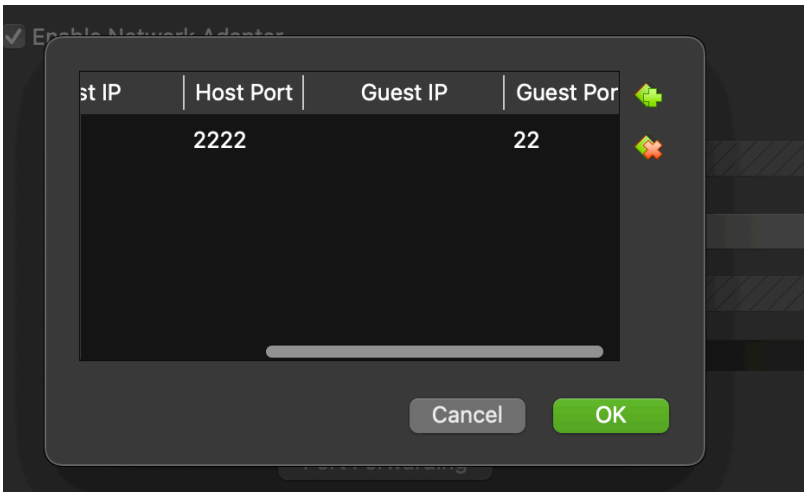
Ip adresu mozemo naci sa naredbom `ip a` ili `hostname -I`

```
vilibald-kovac@vili-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:1b:34:f1 brd ff:ff:ff:ff:ff:ff
    altname enx0800271b34f1
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s8
       valid_lft 85278sec preferred_lft 85278sec
    inet6 fd17:625c:f037:2:a00:27ff:fe1b:34f1/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86100sec preferred_lft 14100sec
    inet6 fe80::a00:27ff:fe1b:34f1/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
vilibald-kovac@vili-server:~$
```

- *Kako ćete provjeriti koji port koristi SSH poslužitelj?* Mozemo vidjeti port u statusu SSH posluzitelja ili pomocu `ss -tuln`

```
vilibald-kovac@vili-server:~$ ss -tuln
Netid       State       Recv-Q      Send-Q          Local Address:Port          Peer Address:Port
udp         UNCONN      0           0                  127.0.0.1:323                0.0.0.0:*
udp         UNCONN      0           0                127.0.0.54:53                  0.0.0.0:*
udp         UNCONN      0           0             127.0.0.53%lo:53                  0.0.0.0:*
udp         UNCONN      0           0          10.0.2.15%enp0s8:68                  0.0.0.0:*
udp         UNCONN      0           0                    [::1]:323                    [::]:*
tcp         LISTEN      0           4096             127.0.0.54:53                  0.0.0.0:*
tcp         LISTEN      0           4096                 0.0.0.0:22                  0.0.0.0:*
tcp         LISTEN      0           4096             127.0.0.53%lo:53                  0.0.0.0:*
tcp         LISTEN      0           4096                    [::]:22                    [::]:*
vilibald-kovac@vili-server:~$ _
```

- Povežite se na SSH poslužitelj putem SSH klijenta na dva načina:
    - korištenjem **NAT adaptera i *port forwardinga***




```
~/ ssh vilibald-kovac@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:Z3FytuLuBnbvcfBUbd2RM5BjBTn9SA1uMYnyiEAAtpU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
vilibald-kovac@127.0.0.1's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic aarch64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat May  9 11:35:34 AM UTC 2026

  System load:             0.0
  Usage of /:              52.2% of 10.21GB
  Memory usage:            17%
  Swap usage:              0%
  Processes:               115
  Users logged in:         0
  IPv4 address for enp0s8: 10.0.2.15
  IPv6 address for enp0s8: fd17:625c:f037:2:a00:27ff:fe1b:34f1

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


vilibald-kovac@vili-server:~$ pwd
/home/vilibald-kovac
vilibald-kovac@vili-server:~$ ls
dobrodosli.txt
vilibald-kovac@vili-server:~$
```

- korištenjem **Bridged adaptera**

```
~/ ssh vilibald-kovac@192.168.1.122
The authenticity of host '192.168.1.122 (192.168.1.122)' can't be established.
ED25519 key fingerprint is SHA256:Z3FytuLuBnbvcfBUbd2RM5BjBTn9SA1uMYnyiEAAtpU.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:5: [127.0.0.1]:2222
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.1.122' (ED25519) to the list of known hosts.
vilibald-kovac@192.168.1.122's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic aarch64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat May  9 11:42:12 AM UTC 2026

  System load:  0.01               Processes:               114
  Usage of /:   52.2% of 10.21GB   Users logged in:         0
  Memory usage: 18%                IPv4 address for enp0s8: 192.168.1.122
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Sat May  9 11:35:34 2026 from 10.0.2.2
vilibald-kovac@vili-server:~$ ls -a
.  ..  .bash_history  .bash_logout  .bashrc  .cache  .lesshst  .profile  .ssh  dobrodosli.txt
vilibald-kovac@vili-server:~$ pwd
/home/vilibald-kovac
vilibald-kovac@vili-server:~$
```

3. Skripta

Putem domaćina, **izradite novu bash skriptu** unutar virtualnog stroja, u direktoriju `/home/username/`. Skripta treba napraviti **detaljan ispis svih datoteka** (uključujući skrivene) iz korijenskog direktorija VM-a (`/`).

Screenshot terminala domacina:

```
vilibald-kovac@vili-server:~$ cat skripta_dz4.sh
#!/bin/bash

echo "Sadrzaj korijenskog direktorija:"

ls -la /
vilibald-kovac@vili-server:~$ ./
.cache/         .config/        skripta_dz4.sh  .ssh/
vilibald-kovac@vili-server:~$ ./skripta_dz4.sh
Sadrzaj korijenskog direktorija:
total 1745996
drwxr-xr-x  20 root root       4096 May  9 10:50 .
drwxr-xr-x  20 root root       4096 May  9 10:50 ..
lrwxrwxrwx   1 root root          7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x   5 root root       4096 May  9 10:51 boot
dr-xr-xr-x   2 root root       4096 Apr 22 09:06 cdrom
drwxr-xr-x  19 root root       3960 May 10 07:45 dev
drwxr-xr-x 109 root root       4096 May  9 11:43 etc
drwxr-xr-x   3 root root       4096 May  9 10:51 home
lrwxrwxrwx   1 root root          7 Apr 20 08:46 lib -> usr/lib
drwx------   2 root root      16384 May  9 10:49 lost+found
drwxr-xr-x   2 root root       4096 Apr 22 08:32 media
drwxr-xr-x   2 root root       4096 Apr 22 08:32 mnt
drwxr-xr-x   2 root root       4096 Apr 22 08:32 opt
dr-xr-xr-x 174 root root          0 May 10 07:45 proc
drwx------   3 root root       4096 Apr 22 08:38 root
drwxr-xr-x  30 root root        860 May 10 07:45 run
lrwxrwxrwx   1 root root          8 Apr 20 08:46 sbin -> usr/sbin
drwxr-xr-x   2 root root       4096 May  9 10:52 snap
drwxr-xr-x   2 root root       4096 Apr 22 08:32 srv
-rw-------   1 root root 1787822080 May  9 10:50 swap.img
dr-xr-xr-x  13 root root          0 May 10 07:46 sys
drwxrwxrwt  11 root root        220 May 10 07:45 tmp
drwxr-xr-x  11 root root       4096 Apr 22 08:32 usr
drwxr-xr-x  13 root root       4096 May  9 10:52 var
vilibald-kovac@vili-server:~$
```

Screenshot iz VM-a:

```
vilibald-kovac@vili-server:~$ pwd
/home/vilibald-kovac
vilibald-kovac@vili-server:~$ ls -a
.  ..  .bash_history  .bash_logout  .bashrc  .cache  .config  .lesshst  .profile  .ssh  dobrodosli.txt  skripta_dz4.sh
vilibald-kovac@vili-server:~$ cat skripta_dz4.sh
#!/bin/bash

echo "Sadrzaj korijenskog direktorija:"

ls -la /
vilibald-kovac@vili-server:~$ ./skripta_dz4.sh
Sadrzaj korijenskog direktorija:
total 1745996
drwxr-xr-x  20 root root       4096 May  9 10:50 .
drwxr-xr-x  20 root root       4096 May  9 10:50 ..
lrwxrwxrwx   1 root root          7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x   5 root root       4096 May  9 10:51 boot
dr-xr-xr-x   2 root root       4096 Apr 22 09:06 cdrom
drwxr-xr-x  19 root root       3960 May 10 07:45 dev
drwxr-xr-x 109 root root       4096 May  9 11:43 etc
drwxr-xr-x   3 root root       4096 May  9 10:51 home
lrwxrwxrwx   1 root root          7 Apr 20 08:46 lib -> usr/lib
drwx------   2 root root      16384 May  9 10:49 lost+found
drwxr-xr-x   2 root root       4096 Apr 22 08:32 media
drwxr-xr-x   2 root root       4096 Apr 22 08:32 mnt
drwxr-xr-x   2 root root       4096 Apr 22 08:32 opt
dr-xr-xr-x 174 root root          0 May 10 07:45 proc
drwx------   3 root root       4096 Apr 22 08:38 root
drwxr-xr-x  30 root root        860 May 10 07:45 run
lrwxrwxrwx   1 root root          8 Apr 20 08:46 sbin -> usr/sbin
drwxr-xr-x   2 root root       4096 May  9 10:52 snap
drwxr-xr-x   2 root root       4096 Apr 22 08:32 srv
-rw-------   1 root root 1787822080 May  9 10:50 swap.img
dr-xr-xr-x  13 root root          0 May 10 07:46 sys
drwxrwxrwt  11 root root        220 May 10 07:45 tmp
drwxr-xr-x  11 root root       4096 Apr 22 08:32 usr
drwxr-xr-x  13 root root       4096 May  9 10:52 var
vilibald-kovac@vili-server:~$
```